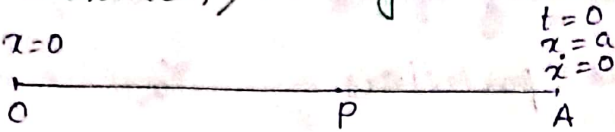


✓ A particle moves in a st. line from a distance a to the centre of force. Show that the time required to reach the centre is $\frac{a^2}{\sqrt{\mu}}$ or $\frac{2a^{4/3}}{\sqrt{3\mu}}$ according as the force of attraction varies inversely as the cube of the distance or as the $\frac{5}{3}$ th power of distance, μ being constant of variation. 1995

Ans. $x=0$



Let O be the centre of force and the particle starts from A moves along AO . such that $AO = a$. Let P be the position of the particle at any time t such that $OP = x$.

Case I $\rightarrow$ Force varies inversely as the cube of the distance.
Equⁿ of the motion of the particle is

$$m v \frac{dv}{dx} = -m \frac{\mu}{x^3}$$

[negative sign taken since the force is directed towards the origin.
 m is the mass of the particle.
 v is the velocity of the particle at P .]

$$\therefore v dv = -\frac{\mu}{x^3} dx$$

Integrating

$$\frac{v^2}{2} = \frac{\mu}{2x^2} + C_1 \rightarrow \text{①} \quad [C_1 = \text{Integrating constant}]$$

Initially when $t=0$, $x=a$ and $\dot{x}=0$

$$C_1 + \frac{\mu}{2a^2} = 0.$$

$$C_1 = -\frac{\mu}{2a^2}$$

From ①

$$\frac{v^2}{2} = \frac{\mu}{2x^2} - \frac{\mu}{2a^2}$$

$$\frac{v^2}{2} = \frac{\mu}{2} \left(\frac{1}{x^2} - \frac{1}{a^2} \right)$$

$$v^2 = \mu \left(\frac{a^2 - x^2}{a^2 x^2} \right)$$

$$v = -\frac{\sqrt{\mu}}{a} \frac{\sqrt{a^2 - x^2}}{x}$$

[Negative sign is taken as t increases, x decreases]

$$\frac{dx}{dt} = -\frac{\sqrt{\mu}}{a} \frac{\sqrt{a^2 - x^2}}{x}$$

$$\frac{\sqrt{\mu}}{a} dt = -\frac{x}{\sqrt{a^2 - x^2}} dx$$

Let t_1 be the time taken by the particle to arrive at origin.
Integrating between the limits $t=0$ when $x=a$ to $t=t_1$ when $x=0$

$$\int_0^{t_1} \frac{\sqrt{\mu}}{a} dt = - \int_a^0 \frac{x dx}{\sqrt{a^2 - x^2}}$$

$$\frac{\sqrt{\mu}}{a} t_1 = \int_0^a \frac{x dx}{\sqrt{a^2 - x^2}}$$

Let $x = a \sin \theta$
 $dx = a \cos \theta d\theta$

$$= \int_0^{\pi/2} \frac{a \sin \theta \cdot a \cos \theta d\theta}{\sqrt{a^2 - a^2 \sin^2 \theta}}$$

$$\begin{array}{c|c|c} x & 0 & a \\ \hline \theta & 0 & \pi/2 \end{array}$$

$$= a \int_0^{\pi/2} \sin \theta d\theta = a [-\cos \theta]_0^{\pi/2}$$

$$= a [-\cos \pi/2 + \cos 0] = a$$

$$\therefore \frac{\sqrt{\mu}}{a} t_1 = a$$

$$t_1 = \frac{a^2}{\sqrt{\mu}} \text{ proved.}$$

Case II $\rightarrow$ Force varies ^{inversely} as the $\frac{5}{3}$ th power of the distance.
 $\therefore$ Equⁿ of the motion of the particle is

$$m v \frac{dv}{dx} = -m \frac{\mu}{x^{5/3}}$$

$$v dv = - \frac{\mu}{x^{5/3}} dx$$

Integrating

$$\frac{v^2}{2} = \left[+ \frac{3\mu}{2x^{2/3}} \right] + C_1 \rightarrow (2) \quad C_1 = \text{Integrating constant.}$$

Initially when $t=0$, $x=a$ and $v=0$

$$\frac{3\mu}{2a^{2/3}} + C_1 = 0$$

$$C_1 = - \frac{3\mu}{2a^{2/3}}$$

From (2)

$$\frac{v^2}{2} = \frac{3\mu}{2x^{2/3}} - \frac{3\mu}{2a^{2/3}}$$

$$v^2 = \frac{3\mu}{x^{2/3}} \left(\frac{1}{a^{2/3}} - \frac{1}{x^{2/3}} \right)$$

$$v^2 = \frac{3\mu}{x^{2/3}} \frac{a^{2/3} - x^{2/3}}{a^{2/3}}$$

$$v = - \sqrt{\frac{3\mu}{x^{2/3}}} \frac{\sqrt{a^{2/3} - x^{2/3}}}{a^{1/3}} \quad \left[\text{Negative sign is taken as } t \text{ increases, } x \text{ decreases} \right]$$

$$\frac{dx}{dt} = -\sqrt{3\mu} \frac{\sqrt{a^{4/3} - x^{4/3}}}{a^{1/3} x^{1/3}}$$

$$\sqrt{3\mu} dt = - \frac{a^{1/3} x^{1/3}}{\sqrt{a^{4/3} - x^{4/3}}} dx \quad * \text{ Put } x^{1/3} = a^{1/3} \sin^2 \theta$$

Let t_2 be the time taken by the particle to arrive at origin. Integrating between the limits $t=0$ to $t=t_2$ to $x=a$ to $x=0$ to

$$\int_0^{t_2} \sqrt{3\mu} dt = - \int_a^0 \frac{a^{1/3} x^{1/3}}{\sqrt{a^{4/3} - x^{4/3}}} dx \quad *$$

$$\sqrt{3\mu} t_2 = \int_0^a \frac{a^{1/3} x^{1/3}}{\sqrt{a^{4/3} - x^{4/3}}} dx$$

$$\sqrt{3\mu} t_2 = \int_0^{\pi/2} \frac{a^{1/3} a^{1/3} \sin^2 \theta \cdot 3a \sin \theta \cos \theta d\theta}{\sqrt{a^{4/3} - a^{4/3} \sin^4 \theta}}$$

$$\sqrt{3\mu} t_2 = \int_0^{\pi/2} 3a^{4/3} \sin^3 \theta d\theta$$

$$\sqrt{3\mu} t_2 = \frac{3a^{4/3}}{4} \int_0^{\pi/2} 4 \sin^3 \theta d\theta$$

$$\sqrt{3\mu} t_2 = \frac{3}{4} a^{4/3} \int_0^{\pi/2} [3 \sin \theta - \sin^3 \theta] d\theta$$

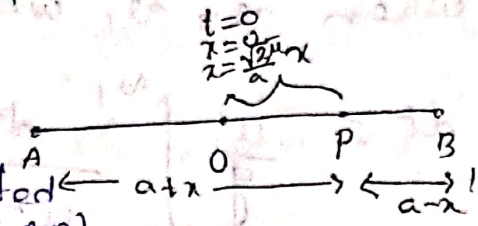
$$\sqrt{3\mu} t_2 = \frac{3}{4} a^{4/3} \left[-3 \cos \theta + \frac{\cos^3 \theta}{3} \right]_0^{\pi/2}$$

$$\sqrt{3\mu} t_2 = \frac{3}{4} a^{4/3} \left[3 - \frac{1}{3} \right] = \frac{3}{4} \times \frac{8}{3} a^{4/3} = 2 a^{4/3}$$

$$t_2 = \frac{2 a^{4/3}}{\sqrt{3\mu}} \quad (\text{proved})$$

Q A particle of unit mass is attracted by two centres of forces A and B each of which attracts with a force $\frac{\mu}{r^3}$ at a distance r . If A, B. AB be $2a$ and the particle be projected from the middle pt. of AB towards B with a velocity $\frac{\sqrt{2\mu}}{a}$, show that it will arrive at B at the end of time $\frac{a^2}{2\sqrt{2\mu}} [3 \log(1+\sqrt{2}) - \sqrt{2}]$

Ans. Let P be the position of the particle at any time t after it is projected from O (the middle pt. of AB) towards B with a velocity $\frac{\sqrt{2\mu}}{a}$. let $OP = x$.



Then the forces of attraction per unit mass to the centres A, B are respectively $\frac{\mu}{(a-x)^3}$ and $\frac{\mu}{(a+x)^3}$ towards A and $\frac{\mu}{(a-x)^3}$ (towards B).

The eqn of motion of the particle per unit mass

$$\frac{d^2x}{dt^2} = \frac{\mu}{(a-x)^3} - \frac{\mu}{(a+x)^3} \quad \text{--- (1)}$$

Multiplying by $2 \frac{dx}{dt}$

$$2 \frac{dx}{dt} \frac{d^2x}{dt^2} = \left[\frac{\mu}{(a-x)^3} - \frac{\mu}{(a+x)^3} \right] \frac{dx}{dt}$$

$$\frac{d}{dt} \left(\frac{dx}{dt} \right)^2 = 2\mu \left[\frac{1}{(a-x)^3} - \frac{1}{(a+x)^3} \right] \frac{dx}{dt}$$

Integrating

$$\left(\frac{dx}{dt} \right)^2 = 2\mu \left[\frac{1}{2(a-x)^2} + \frac{1}{2(a+x)^2} \right] + C$$

[C = const]

Initially $t=0$, $x=0$, $\dot{x} = \frac{\sqrt{2\mu}}{a}$

$$\frac{2\mu}{a^2} = 2\mu \left[\frac{1}{a^2} + \frac{1}{a^2} \right] + C_1$$

$$\frac{2\mu}{a^2} = \frac{2\mu}{a^2} + C_1$$

$$C_1 = 0.$$

$$\therefore \left(\frac{dx}{dt} \right)^2 = 2\mu \left[\frac{1}{2(a-x)^2} + \frac{1}{2(a+x)^2} \right]$$

$$\text{or, } \left(\frac{dx}{dt} \right)^2 = \frac{2\mu(a^2 + x^2)}{(a^2 - x^2)^2}$$

$$\therefore \frac{dx}{dt} = \frac{\sqrt{2\mu} \sqrt{a^2 + x^2}}{a^2 - x^2} \quad \left[\text{we take positive sign since as } t \text{ increases, } x \text{ increases} \right]$$

Let t_1 be the time taken by the particle to reach the pt. B.

Integrating between the limits, $t=0$ when $x=0$ to $t=t_1$ when $x=a$ we get from (2) -

$$\sqrt{2\mu} \int_0^{t_1} dt = \int_0^a \frac{a^2 - x^2}{\sqrt{a^2 + x^2}} dx$$

$$\text{or, } \sqrt{2\mu} t_1 = \int_0^a \frac{a^2 - x^2}{\sqrt{a^2 + x^2}} dx$$

$$= a^2 \int_0^a \frac{1}{\sqrt{a^2 + x^2}} dx - \int_0^a \frac{x^2}{\sqrt{a^2 + x^2}} dx$$

$$= a^2 \left[\int_0^a \frac{1}{\sqrt{a^2 + x^2}} dx - \int_0^a \frac{x^2}{\sqrt{a^2 + x^2}} dx \right]$$

$$= a^2 \left[\int_0^{\pi/4} \sec \theta d\theta - \int_0^{\pi/4} (\sec^2 \theta - 1) \sec \theta d\theta \right]$$

$$= a^2 \left[\int_0^{\pi/4} \sec \theta d\theta - \int_0^{\pi/4} \sec^3 \theta d\theta + \int_0^{\pi/4} \sec \theta d\theta \right]$$

$$= a^2 \left[2 \int_0^{\pi/4} \sec \theta d\theta - \int_0^{\pi/4} \sec^3 \theta d\theta \right]$$

$$= a^2 \left[2 \left[\ln |\sec \theta + \tan \theta| \right]_0^{\pi/4} - \left[\frac{\sec \theta \tan \theta}{2} + \frac{\ln |\sec \theta + \tan \theta|}{2} \right]_0^{\pi/4} \right]$$

$$= a^2 \left[2 \left[\ln |\sec \frac{\pi}{4} + \tan \frac{\pi}{4}| \right] - \left[\frac{\sec \frac{\pi}{4} \tan \frac{\pi}{4}}{2} + \frac{\ln |\sec \frac{\pi}{4} + \tan \frac{\pi}{4}|}{2} \right] \right]$$

$$= a^2 \left[2 \left[\ln |\sqrt{2} + 1| \right] - \left[\frac{1}{2} + \frac{\ln |\sqrt{2} + 1|}{2} \right] \right]$$

$$= a^2 \left[\ln |\sqrt{2} + 1| - \frac{1}{2} \right]$$

$$\sqrt{2\mu} \, h = a^2 \int_0^{\pi/4} \frac{\sin \theta \, d\theta}{1 + \tan^2 \theta} = a^2 \int_0^{\pi/4} \frac{\sec^3 \theta}{\sec^2 \theta} \, d(\tan \theta)$$

$$= a^2 \int_0^{\pi/4} 2 \log(\sec \theta + \tan \theta) \, d\theta$$

$$\sec^3 \theta = \sec^2 \theta \cdot \sec \theta = \sqrt{1 + \tan^2 \theta} \, d(\tan \theta)$$

$$\sqrt{2\mu} \, h = \int_0^a \frac{2a^2 - (a^2 + x^2)}{\sqrt{a^2 + x^2}} \, dx$$

$$= 2a^2 \int_0^a \frac{dx}{\sqrt{a^2 + x^2}} - \int_0^a \frac{x \, dx}{\sqrt{a^2 + x^2}}$$

$$= 2a^2 \left[\log(x + \sqrt{x^2 + a^2}) \right]_0^a - \left[\frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log|x + \sqrt{x^2 + a^2}| \right]_0^a$$

$$= 2a^2 \left[\log a(1 + \sqrt{2}) - \log a \right] - \left[\frac{a}{2} \sqrt{2}a + \frac{a^2}{2} \log(1 + \sqrt{2}) \right]$$

$$= 2a^2 \log(1 + \sqrt{2}) - \frac{a^2}{\sqrt{2}} - \frac{a^2}{2} \log(1 + \sqrt{2})$$

$$= \frac{3a^2}{2} \log(1 + \sqrt{2}) - \frac{a^2}{\sqrt{2}}$$

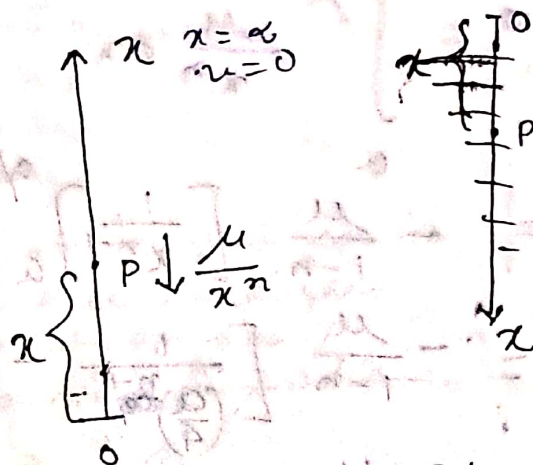
$$\sqrt{2\mu} \, h = \frac{a^2}{2} \left[3 \log(1 + \sqrt{2}) - \sqrt{2} \right]$$

$$h = \frac{a^2}{2\sqrt{2}\mu} \left[3 \log(1 + \sqrt{2}) - \sqrt{2} \right]$$

This is the required time when it arrives at B.

Q. A particle is attracted by a force to a fixed pt. varying inversely as (distance)ⁿ. If the velocity acquired in falling from an infinite distance to a distance a from the centre be equal to the velocity acquired in falling from rest at a distance a to a distance $\frac{a}{4}$, then show that $n = 3/2$

Ans.



Let O be the centre of force and P be the position of the particle at any time t.

Equⁿ of motion of the particle is

$$m\ddot{x} = -\frac{\mu}{x^n}$$

$$\ddot{x} = -\frac{\mu}{x^n} \quad \text{--- (1)}$$

Equⁿ (1) can be written as

$$v \frac{dv}{dx} = -\frac{\mu}{x^n}$$

$$v dv = -\frac{\mu}{x^n} dx \quad \text{--- (2)}$$

Let the particle acquires a velocity v_1 in falling from an infinite distance to a distance a .

Integrating (2) between the limits $x = \infty$ when $v = 0$ to $x = a$ when $v = v_1$, we get

$$\int_0^{v_1} v dv = -\int_{\infty}^a \frac{\mu}{x^n} dx$$

$$\frac{v_1^2}{2} = -\frac{\mu}{1-n} \left[\frac{1}{x^{n-1}} \right]_{\infty}^a$$

$$\therefore \frac{v_1^2}{2} = -\frac{\mu}{1-n} \frac{1}{a^{n-1}} \quad n > 1$$

$$= \frac{\mu}{n-1} \frac{1}{a^{n-1}}$$

Let v_2 be the velocity acquired by the particle in falling from a distance a to a distance $a/4$.

Integrating between the limits $x=a$ when $v=0$ when $x=a/4$ when $v=v_2$ we get

$$\int_0^{v_2} v dv = \int_a^{a/4} \frac{\mu}{x^n} dx$$

$$\frac{v_2^2}{2} = -\frac{\mu}{1-n} \left[\frac{1}{x^{n-1}} \right]_a^{a/4}$$

$$= -\frac{\mu}{1-n} \left[\frac{1}{(a/4)^{n-1}} - \frac{1}{a^{n-1}} \right]$$

$$= \frac{\mu}{n-1} \left[\frac{4^{n-1} - 1}{a^{n-1}} \right]$$

According to the problem $v_1 = v_2$

$$\frac{\mu}{(n-1)a^{n-1}} = \frac{\mu}{n-1} \frac{4^{n-1} - 1}{a^{n-1}}$$

$$1 = 4^{n-1} - 1$$

$$4^{n-1} = 2^2$$

$$2n-2 = 2$$

$$\boxed{n = 3/2} \text{ (proved)}$$